

Automated Depression Detection From Text and Audio: A Systematic Review

Yuxin Li , Sinchana Kumbale, Yanru Chen , Tanmay Surana, Eng Siong Chng , *Senior Member, IEEE*, and Cuntai Guan , *Fellow, IEEE*

Abstract—Depression is a prevalent mental health disorder that presents significant challenges for timely diagnosis and intervention. Automated Depression Detection (ADD) systems using text and audio offer scalable mental health assessment solutions. This review systematically evaluates 65 studies published between 2018 and 2024, focusing on ADD methods that utilize machine learning models with multimodal data. We examine key methodologies, including data augmentation, multimodal fusion, and feature extraction, along with state-of-the-art ADD systems. The review emphasizes the need for culturally adaptable, high-quality datasets and interpretable models for clinical use. We also identify gaps in longitudinal data and real-world applications. Future research should focus on developing clinically integrated, cross-cultural ADD systems that are interpretable, scalable, and robust. The findings of this review contribute to the research field by providing a comprehensive overview of existing methodologies, identifying gaps in the current literature, and offering insights for future advancements in depression detection using speech and text analysis.

Index Terms—Biomedical signal processing, deep learning, mental health technology, health informatics.

I. INTRODUCTION

DEPRESSION is a pervasive mental health disorder characterized by persistent feelings of sadness, loss of interest or pleasure, and a range of cognitive and physical symptoms. The World Health Organization (WHO) identifies depression as one of the leading causes of disability worldwide, affecting an estimated 280 million people, with over 700,000 annual deaths attributed to suicide [1]. Accurate and early detection of depression is essential for timely intervention and improving patient outcomes. Traditionally, depression diagnosis relies on clinical interviews and self-reporting tools such as the Beck Depression Inventory (BDI-II) [2] and the Patient Health Questionnaire-9

(PHQ-9) [3]. These traditional methods heavily depend on patient self-reporting, which is subject to potential bias, and are further limited by a shortage of mental health professionals. These factors highlight the need for more efficient, scalable, and objective diagnostic tools.

In recent years, advances in machine learning (ML) and deep learning (DL) have facilitated the development of Automated Depression Detection (ADD) systems, which aim to complement traditional methods by leveraging digital biomarkers from text and audio data. These modalities are particularly advantageous as they are readily available, and provide rich, complementary insights into a person's mental state [4], [5], [6], [7], [8], [9]. These technologies have shown promise in identifying depressive symptoms and are highly accessible for both research and clinical use.

Previous review papers and surveys have summarized depression detection methods through various data modalities, including audio [10], [11], [12], text [13], [14], [15], visual cues [16], EEG signals [17], [18], and multimodal fusion (text, audio, and visual) [19]. However, a comprehensive analysis focusing on the synergistic use of text and audio modalities, particularly in light of recent advances in deep learning and large language models, remains needed.

This systematic review aims to provide an in-depth analysis of state-of-the-art ADD methods based on text and audio data. It addresses several key questions related to the use of ADD systems.

- 1) Which features are extracted and which techniques are used in current ADD studies to support clinical depression diagnosis?
 - 2) How effective are current machine learning methods in detecting depression from diverse data sources?
 - 3) What strategies have been proposed to address the issue of data scarcity in ADD?
 - 4) What are the strengths and limitations of recent text, audio, and bi-modal ADD methods for scalable mental health assessment?
- What are the primary challenges associated with implementing text/audio-based ADD methods in clinical systems?
- 5) How can future advancements improve the accuracy, interpretability, and accessibility of ADD methods for mental health diagnostics?

We systematically review studies published between 2018 and 2024 to highlight the key methodological advances, including the integration of multimodal data, the use of data augmentation

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Yuxin Li, Yanru Chen, Tanmay Surana, Eng Siong Chng, and Cuntai Guan are with the College of Computing and Data Science, Nanyang Technological University, Singapore 639798 (e-mail: yuxin.li@ntu.edu.sg; ch0009ru@e.ntu.edu.sg; tanmay.surana@ntu.edu.sg; aseschng@ntu.edu.sg; CTGuan@ntu.edu.sg).

Sinchana Kumbale is with the Department of Computer Science and Engineering, Manipal Institute of Technology, Manipal 576104, India (e-mail: sinchana.kumbale@learner.manipal.edu).

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techniques, and the challenges of model interpretability and clinical implementation. By focusing on these critical areas, this paper seeks to advance the development of ADD systems that are scalable, interpretable, and clinically applicable across diverse cultural settings. The findings and insights presented here are intended to guide future research and foster the integration of ADD tools into healthcare systems, contributing to more accessible, accurate, and timely mental health diagnostics.

The structure of the review paper is organized as follows. In Section II, we introduce the background of Major Depressive Disorder (MDD) diagnosis, along with supporting evidence for using text and audio as biomarkers for detecting depression. Section III details the methodology of this review, including the search strategy, selection process, and data extraction methods used. Section IV covers the synthesis of findings, including public depression datasets, data augmentation methods, feature engineering methods, and the existing depression detection systems. Section V discusses the limitation and future direction associated with ADD. Section VI concludes the main findings in this review.

II. BACKGROUND OF MAJOR DEPRESSIVE DISORDER DIAGNOSIS

A. The Definition of Depression

Clinical depression, also known as Major Depressive Disorder (MDD), is a significant medical illness responsible for considerable morbidity and disability. As defined by the World Health Organization (WHO), it is a multifaceted mental health disorder characterized by persistent feelings of sadness, emptiness, or hopelessness, often accompanied by a loss of interest or pleasure in activities previously enjoyed. Typically, these symptoms must persist for at least two weeks and significantly interfere with daily functioning, leading depression to become a leading cause of disability worldwide [1]. Globally, an estimated 3.8% of the population experiences depression, equating to approximately 280 million people, with over 700,000 annual deaths attributed to suicide—a critical public health issue particularly affecting individuals aged 15 to 29 [20].

Neuroscientific research supports the biopsychosocial model, indicating that depression results from complex interactions among biological, psychological, and social factors. Biological evidence includes neurochemical imbalances, dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis, immune system dysfunction, and structural and functional brain abnormalities, particularly in regions like the hippocampus and prefrontal cortex. These disruptions correlate with cognitive impairments, such as deficits in attention, memory, and emotional processing [2], highlighting the necessity for multimodal detection methods.

Overall, depression represents a significant public health challenge that requires an integrated understanding across different levels of neuroscientific research for effective management and treatment.

B. Traditional Clinical Diagnosis of Major Depression Disorder

The traditional diagnosis of Major Depression Disorder (MDD) is primarily conducted through clinical evaluations by psychiatrists or by completing self-report questionnaires based on criteria from the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) [21]. Both methods require the presence of at least five specific symptoms during a two-week period, with at least one being either a depressed mood or a loss of interest or pleasure [21]. Popular psychological assessments include Beck Depression Inventory (BDI-II) [2], (Patient Health Questionnaire) PHQ-8 [22], PHQ-9 [3], PHQ-15 [23], Quick Inventory of Depressive Symptoms (QIDS) [24], Inventory of Depressive Symptomatology (IDS) [25], Zung Self-Rating Depression Scale (Zung-SDS) [26], Geriatric Depression Scale (GDS) [27], Major Depression Inventory (MDI) [28], Mood and Feelings Questionnaire (MFQ) [29], Clinically Useful Depression Outcome Scale (CUDOS) [30], Carroll Rating Scales (CRS) [31], (HAM-D) [32], Montgomery-Asberg Depression Scale (MADRS) [33], IDS [25] and so on. The HAM-D [32] is the most commonly used in traditional clinical diagnosis. The different diagnostics scales have different questions and ranges of scores for the level of depression, and Table I details the information for each scale.

The main issue for traditional diagnostic method is the shortage of psychiatrists, which means there aren't enough professionals to provide accurate and timely depression diagnoses. Within a few years, the United States is projected to be short between 14,280 and 31,109 psychiatrists, while psychologists, social workers, and other mental health professionals will also be overextended [34]. Additionally, the gap between need and access is particularly pronounced among certain populations, including those in rural areas [35]. Compounding the issue, more than 60% of psychiatrists are 55 or older, leading to a retirement drain that further strains the mental health care system [36]. Another limitation is the potential stigma when self-reporting. Studies have shown that individuals may be reluctant to disclose their depressive symptoms due to fear of judgment, societal stigma, or concerns about confidentiality [37]. The American Psychological Association highlights that self-reporting bias can lead to underreporting of symptoms, resulting in inadequate diagnosis and treatment [38]. Therefore, while traditional methods for diagnosing MDD are foundational and widely used, their limitations highlight the need for more objective, efficient, and accessible diagnostic tools. Therefore, an increasing number of studies are exploring automated methods using machine learning and deep learning approaches, which hold promise for addressing these challenges and enhancing the accuracy and accessibility of depression diagnosis.

C. Text and Audio as Biomarkers for Depression

Text and audio data have increasingly been recognized as valuable biomarkers for detecting depression, supported by psychological and linguistic research. These modalities provide scalable means of assessing mental health and have been shown

TABLE I
DETAILS ABOUT DEPRESSION DIAGNOSTIC SCALES

Scale	Year	Normal	Mild	Moderate	Severe	Very Severe	Type
HAM-D	1986	0-7	8-13	14-18	19-22	23-53	Interview
BDI-II	1996	0-13	14-19	20-28	29-63	—	Self-report
PHQ-8	2009	0-4	5-9	10-14	15-19	20-24	Self-report
PHQ-9	2001	0-4	5-9	10-14	15-19	20-27	Self-report
PHQ-15	2002	1-4	5-9	10-14	15-30	—	Self-report
QIDS	2003	0-5	6-10	11-15	16-20	21-48	Self-report
MADRS	1979	0-11	12-22	23-30	31-35	36-60	Interview
IDS	1996	0-13	14-25	26-38	39-48	49-90	Interview & Self-report
Zung-SDS	1965	≤49	50-59	60-69		70-80	Self-report
CUDOS	2008	0-10	11-20	21-30	31-45	46-72	Self-report
MDI	2012	0-19	20-14	25-29		30-50	Self-report
MFQ	1988	-	-	-	-	-	Self-report
GDS	1982	0-4	5-8	9-11	12-15	-	Self-report
CRS	1981	-	-	-	-	-	Self-report

Note: "Normal", "Mild", "Moderate", "Severe", "Very Severe" refer to scores from the scales.

to reveal cognitive and emotional states related to depressive disorders.

Textual data, derived from spoken or written content, contains critical markers that reflect a person's psychological state. Research has shown that individuals with depression exhibit distinct linguistic patterns [4] [5] [6]. These patterns can be systematically extracted and analyzed through natural language processing (NLP) techniques.

Similarly, audio data, particularly speech, provides rich acoustic information that correlates with emotional and cognitive disruptions common in depression [7]. Changes in prosodic features such as tone, pitch, speech rate, and pause length are often observed in individuals suffering from depression. These speech characteristics reflect underlying emotional dysregulation and cognitive fatigue, which are hallmark symptoms of the disorder [9], [39], [40], [41], [42]. Automated systems can process these audio features to detect depressive symptoms, making audio an effective modality for passive mental health monitoring.

The combination of text and audio enhances the accuracy of depression detection by offering complementary insights. Text reflects cognitive distortions and emotional distress, while audio captures vocal expressions of emotional tone. Together, they provide a more comprehensive view of a person's mental state, aligning with recent advances in ADD systems that leverage both modalities for scalable and objective assessments.

III. METHODOLOGY OF THE REVIEW

In this section, we outline the methods employed to conduct a systematic review on ADD from text and audio. We followed the guidelines from PRISMA [43] to identify studies. Our methodology encompasses a detailed search strategy, a structured selection process and data extraction protocol to ensure the inclusion of relevant and high-quality studies. Finally, we present a structured classification scheme used in this review paper. This approach allows for a thorough synthesis of existing research,

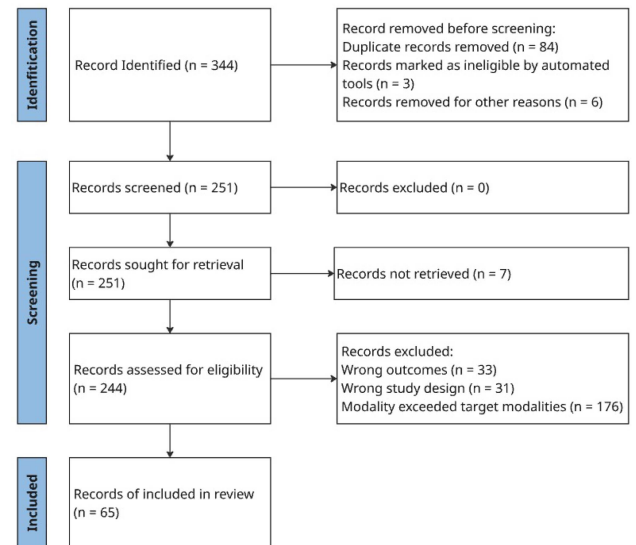


Fig. 1. Identifications of studies via databases.

enabling a deeper understanding of current advancements and challenges in the field. The flowcharts is presented in Fig. 1.

A. Search Strategy

The literature search was conducted across multiple databases to ensure comprehensive coverage of the relevant research. We searched major academic databases including Google Scholar, PubMed, IEEE Xplore and other prominent scientific repositories [44], [45], [46], [47], [48], [49], [50], [51], [52]. Additionally, sources such as conference proceedings, relevant journals, and institutional repositories were included to gather a wide range of studies on ADD from text and audio.

The search strategy employed a combination of specific keywords and Boolean operators to refine the search results. The main keywords used included “automated depression detection”, “text-based depression detection”, “audio-based depression detection”, “speech-based depression detection”, “major

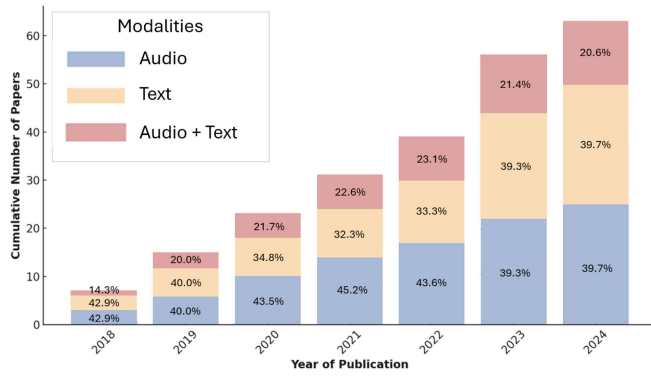


Fig. 2. Modality distribution across years.

depression disorder assessment”, “machine learning”, “deep learning” and “multi-modal depression detection.” Boolean operators such as AND, OR, and NOT were utilized to combine and exclude certain terms to ensure a focused search. For example, searches like “depression detection AND text” and “depression detection AND audio” were conducted to gather targeted studies.

The inclusion criteria for selecting articles were as follows: (1) Articles published from 2016 to 2024 were considered to ensure the inclusion of recent advancements in the field. (2) Only articles published in English were included. (3) Empirical studies and meta-analyses focusing on ADD from text and audio were included. (4) Studies from disciplines such as computer science, psychology, psychiatry, and linguistics were considered relevant.

B. Selection Process and Data Extraction

The initial screening involved a two-step process. First, the titles and abstracts of the articles retrieved from the databases were screened for relevance. Articles that did not meet the inclusion criteria based on their titles and abstracts were excluded at this stage. For articles that passed the initial screening, the full texts were reviewed to ensure they met the inclusion criteria. This involved a thorough examination of the study’s objectives, methodology, results, and relevance to ADD from text and audio.

A total of 251 studies were initially identified through the database searches. After the title and abstract screening, 98 articles were selected for full-text review. Of these, 65 studies met all the inclusion criteria and were included in the final review. The distribution of different modalities (Audio, Text, and Audio + Text) used in depression detection research from 2016 to 2024 is shown in Fig. 2. The primary reasons for excluding articles at the full-text review stage included irrelevance to the topic, lack of methodological rigor, and focus on non-automated methods.

From these selected studies, data extraction was conducted to summarize key findings, themes, and methodological approaches from the included studies. Information extracted included study objectives, methods used for ADD (text, audio, or both), sample sizes, data sources, types of machine learning models employed, performance metrics, and key results.

C. Classification Scheme

We adopted a structured classification scheme to review and compare existing ADD methods. As outlined in the taxonomy diagram in Fig. 3, each methodological component—data modality, feature representation, model architecture, training strategy, fusion mechanism, and evaluation metric—was assessed not only by its structural characteristics but also by its practical and empirical contributions. Specifically, a study was included if its methodological improvement met one or more of the following criteria: (1) Introduced novel feature extraction techniques; (2) Proposed new machine/deep learning architectures tailored for depression detection, particularly those addressing temporal dependencies or multimodality; (3) Addressed data scarcity through techniques like augmentation (e.g., generative adversarial networks for synthetic data), transfer learning, or cross-dataset validation. (4) Developed novel fusion mechanisms to integrate text and audio modalities, enhancing robustness over unimodal approaches; (5) Demonstrated statistically significant improvements over existing baselines.

IV. SYNTHESIS OF FINDINGS

This section presents the findings of our review, which explores various data sources for text and audio and then examines the data augmentation methods applied to address data scarcity in this research field. Additionally, we discuss different feature engineering techniques used across text and audio. Most importantly, we provide an organized summary of the different models used as depression classifiers, highlighting their development trajectories and predicting future trends. Through the review, the flowchart of standard methodological steps in ADD systems using unimodal or bimodal is presented in Fig. 4.

A. Depression Detection Datasets

The major types of datasets used in relevant studies are social media data and conversational data. Among the papers reviewed in our study, 44.4% utilize social media data, while 55.6% rely on conversational data. This section demonstrates and analyzes commonly used datasets from both sources.

1) *Social Media-Based Datasets*: The datasets sourced from social media provide rich resources of depression-related text. They often rely on platforms such as X (formerly Twitter) [53], Reddit [54], Facebook [55], and YouTube [56]. These platforms provide a rich repository of user-generated content, which can be leveraged to analyze various aspects of human behavior, sentiment, and interactions in the context of mental health studies.^{1,2} The depression datasets sourced from social media platforms often crawl data with depression-related features via APIs, then refine this data using specific words, phrases, or required hashtags. Examples of such keywords include “depression,” “diagnosis,” and other terms that have been identified by prior research. Sometimes, user behaviors and users’ online timelines are analyzed [57]. While social media provides a rich source of data with a vast amount of information,

¹ <https://www.statista.com/forecasts/1146722/twitter-users-in-the-world>

² <https://www.statista.com/statistics/1310710/redditcom-monthly-users/>

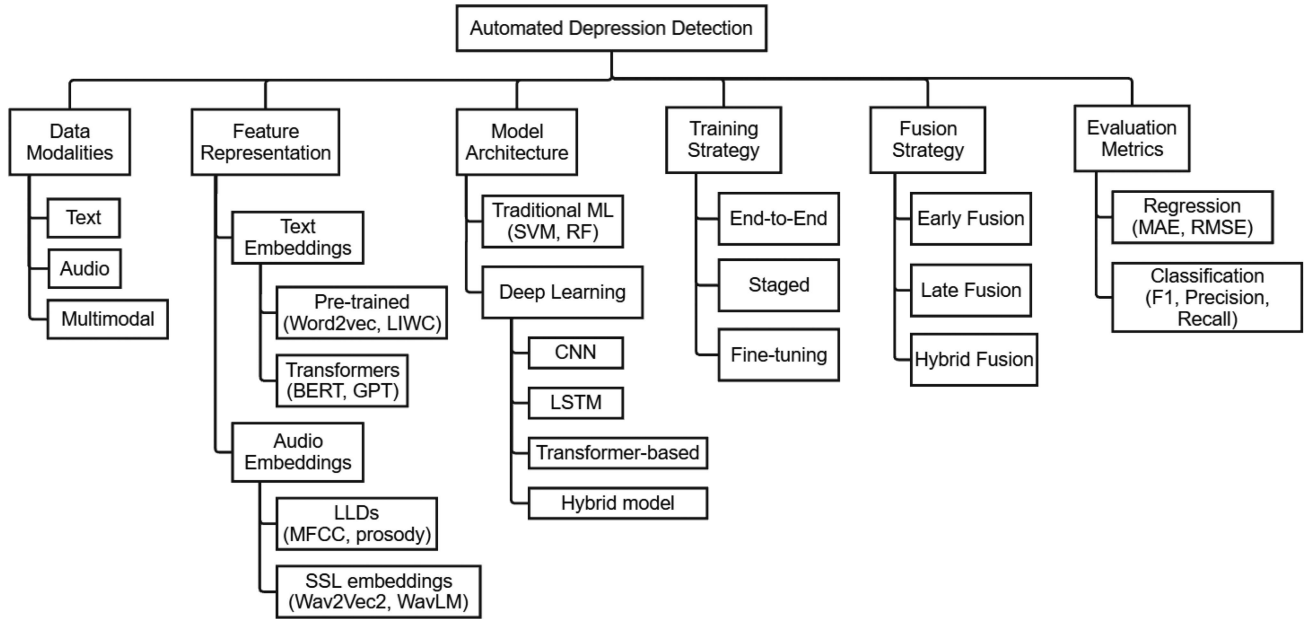


Fig. 3. Taxonomy diagram.

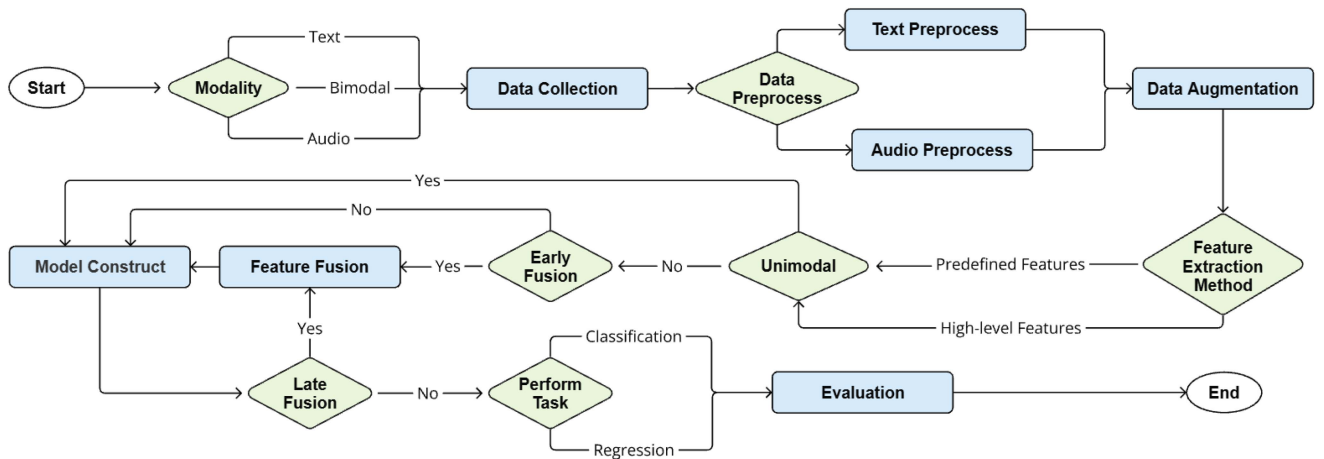


Fig. 4. Flowchart of standard methodological steps in ADD systems using unimodal or bimodal data.

there are concerns regarding the reliability of the data and annotations, which might be biased because the depression labels are decided from just one tweet instead of a diagnostic questionnaire [58].

Due to varying data collection methods, datasets from the same platform often differ and most are not publicly available. Details like modalities, number of records and social media platforms of some publicly available social media-based datasets are summarized in Table II. Several publicly available datasets, including DEPTWEET [59], eRISK [60], RSDD [61], D-vlog [62], SWDD [63], Depression Reddit Dataset [64] and the Depression Severity Dataset [65], have been annotated using different procedures. In the DEPTWEET dataset, each tweet was annotated by at least three annotators, with the label being one of four categories: non-depressed, mildly depressed, moderately

TABLE II
OVERVIEW OF PUBLIC SOCIAL MEDIA DATASETS FOR ADD

Dataset Name	Language	Modalities	Records No.	Platforms
eRISK [66]	English	Text	1304	Reddit
DEPTWEET [67]	English	Text	40191	Twitter
SWDD [63]	Chinese	Text	4854421	Weibo
RSDD [61]	English	Text	598	Reddit
Depression Reddit Dataset [64]	English	Text	-	Reddit
Depression Severity Dataset [65]	English	Text	3553	Reddit
D-vlog [62]	English	Audio and Visual	961	YouTube

depressed, or severely depressed. The final label was determined by majority vote, and if none of the three annotators agreed, the tweet was discarded. For SWDD dataset, eRISK, RSDD, and the Depression Severity Dataset, researchers initially categorized users based on their overall behavioral and emotional patterns, labeling possible depressive symptoms (e.g., non-depressed,

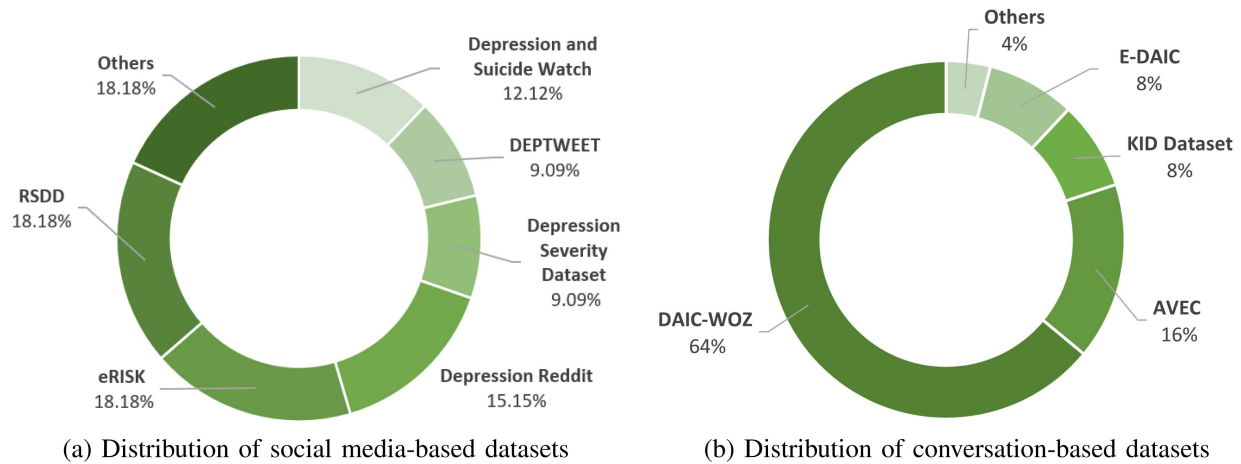


Fig. 5. Distribution of datasets used across reviewed research. Note: The percentage indicates the percentage of papers that utilise each dataset.

mildly depressed, moderately depressed, severely depressed). These labels were then reviewed by a psychologist for accuracy. In the D-vlog dataset, videos were first converted to text and labeled using NLP tools. These labels were then manually reviewed by mental health experts. The distribution of some commonly used social media datasets is displayed in Fig. 5(a).

2) Conversation-Based Datasets: Conversation-based datasets refer to datasets containing conversations between an interviewer and a participant. In recent years, these types of datasets have been widely used due to their advancements in using audio as a cue to detect depression, covering comprehensive depression-related topics, providing rich content, and presenting patient characteristics effectively. These datasets are particularly popular among methods employing multi-modality to detect depression. However, conversation-based datasets face the limitation of being low-resource due to the confidential nature of the data.

Conversation-based datasets often include audio recordings, and sometimes transcripts and audio features of conversations conducted with various participants. The format of these conversations can vary; interviews may be free-form or include a set of predefined questions. In some datasets, participants are asked to perform reading tasks. Unlike social media-based data, where participant labels are largely derived from manual annotation, most conversation-based datasets utilize questionnaires based on diagnostic scales to determine participant labels. Below are brief introductions and availability information for frequently used conversation-based datasets. Table I presents the modalities, language, number of recordings, and the scale for labels provided by each public dataset.

- **DAIC-WOZ [68]:** The DAIC-WOZ dataset consists of interviews conducted by a virtual agent named Ellie, who is controlled by a human interviewer. The dataset comprises of audio recordings and audio features, manually annotated transcripts, and video features, and provides labels based on PHQ-8. This dataset is publicly available on request through the official website [88].
- **E-DAIC [69]:** E-DAIC is an extended version of the DAIC-WOZ dataset. While a large number of factors are similar

to the DAIC-WOZ dataset, one of the key differences is that E-DAIC also contains fully automated interviews with Ellie, and the provided transcripts only contain the participant responses, while DAIC-WOZ includes questions by interviewers as well. The dataset includes audio, video, and text data alongside PHQ-8-based labels. It should be noted that the transcripts in E-DAIC are not manually annotated but generated by an automatic speech recognition (ASR) system. This dataset is publicly available on request through the official website [88].

- **CMDC [70]:** The Chinese Multimodal Depression Corpus (CMDC) is a specialized dataset designed to aid in the study of depression using multimodal data, focusing on native Chinese speakers. It includes recordings of speech, facial expressions, and textual responses from participants, all aimed at identifying depressive tendencies. The data were gathered through clinically validated semi-structured interviews. To create the ground truth labels, participants were evaluated using the HAMD-17 and PHQ-9 scales. This dataset is publicly available [70].
- **AVEC 2013-14 [71], [72]:** These datasets are subsets of a larger corpus called AVID [89]. The specific tasks chosen were reading a short passage (*North Wind and the Sun*³ or *Grandfather*)⁴ and participating in a free-form interview. The dataset consists of video and audio data and labels based on the BDI scores. While the official website created for access to the dataset is no longer accessible, the dataset is publicly accessible on request to the authors.
- **The Androids Corpus [73]:** The Androids Corpus comprises 228 recordings from 118 native Italian speakers, categorized into 64 diagnosed with depression and 54 controls. It includes both read and spontaneous speech, providing a total of nearly 9 hours of audio data. The corpus is structured to allow for both reproducibility of

³ <https://read.gov/aesop/143.html>

⁴ <https://eatspeakthink.com/wp-content/uploads/2020/04/Grandfather-passage.pdf>

TABLE III
OVERVIEW OF PUBLIC CONVERSATIONAL DATASETS FOR ADD

Dataset Name	Language	Modalities	Subjects No.	Patients No.	Label
DAIC-WOZ [68]	English	Audio, Visual and Text	189	56	PHQ 8
E-DAIC [69]	English	Audio, Visual and Text	219	49	PHQ 8
CMDC [70]	Chinese	Audio, Visual and Text	78	26	PHQ-9, HAMD-17
AVEC 2013 [71]	German	Audio and Visual	292	-	BDI II
AVEC 2014 [72]	German	Audio and Visual	84	34	BDI II
The Androids Corpus [73]	Italian	Audio	116	64	PHQ-9, CTQ-SF, LES, SSRS, GAD-7, PSQI
MODMA [74]	Chinese	Audio and EEG	78	26	SDS
EATD [75]	Chinese	Audio and Text	162	30	

TABLE IV
OVERVIEW OF DATA AUGMENTATION METHODS APPLIED IN ADD STUDIES

Data Augmentation Methods	Modalities	Studies	Dataset	Before	After	Impr (%)
Easy Data Augmentation	Text	Ansari et al., [76]	Dreaddit SDCNL	F1: 0.73 F1: 0.74	F1: 0.75 F1: 0.75	2.73↑ 1.35↑
Back Translation	Text	Ansari et al., [76]	Dreaddit SDCNL	F1: 0.73 F1: 0.74	F1: 0.74 F1: 0.74	1.36↑ 0
Synthetic Data Generation	Text	Wang et al., [77] Kang et al., [78] Chen et al., [79]	— DAIC-WOZ DAIC-WOZ E-DAIC	— RMSE: 4.71 F1: 0.8148 Fq: 0.7318	— RMSE: 3.66 F1: 0.8462 F1: 0.7531	— 22.08↓ 3.85↑ 2.91↑
Conditional Data Generation	Text	Ansari et al., [76]	Dreaddit SDCNL	F1: 0.73 F1: 0.74	F1: 0.76 F1: 0.75	4.10↑ 1.35↑
Traditional Audio Augmentation	Speech	Othmani et al., [80] Rejaibi et al., [81]	DAIC-WOZ DAIC-WOZ	— Acc: 0.70	F1: 0.49 Acc: 0.74	— 6.39↑
Test Time Augmentation	Speech	Chelasta et al., [82]	DAIC-WOZ	Acc: 0.71	Acc: 0.77	8.45↑
Fraug	Speech	Ravi et al., [83]	DAIC-WOZ CONVERGE	F1: 0.619 F1: 0.654	0.656 0.739	5.97↑ 12.99↑
SpecAugment	Speech	Kim et al., [84]	Self-collected	—	F1 0.77	—
Topic-based Augmentation	Text Text + Speech	Lam et al., [85]	DAIC-WOZ	F1: 0.71 F1: 0.69	F1: 0.78 F1: 0.78	9.85↑ 13.04↑
Sub-Dialogue Shuffling [84]	Speech Text	Wu et al., [86] Zhang et al., [87]	DAIC-WOZ DAIC-WOZ	F1: 0.451 —	F1: 0.669 F1: 0.719	48.33↑ —

research and comparative analysis, which are essential for developing and refining computational models for mental health diagnostics. A key feature of this dataset is its adherence to rigorous clinical and scientific standards, including the use of DSM-5 for diagnosing depression among participants. Statistical analysis confirms no significant demographic differences (age, gender, education) between the control and depressed groups, ensuring that speech variations are attributed to depressive states rather than other variables. This dataset is publicly available [90].

- **MODMA [74]:** The MODMA dataset contains audio recordings, EEG features and corresponding labels, such as PHQ-9 and QIDS. Each recording consists of three tasks: an interview comprising 18 questions, reading a passage, and describing a picture. The dataset is available on request through the official website [91].
- **EATD [75]:** The EATD-Corpus is a dataset consists of audio and text files of 162 volunteers who received counseling. The depression label is based on the Zung-SDS score. This dataset is publicly available.⁵

Three of the most prominent datasets for ADD are DAIC-WOZ, AVEC 2013 and AVEC 2014 datasets. The distribution of some commonly used conversation-based datasets are displayed through Fig. 5(b).

B. Data Augmentation Methods Used in ADD Systems

To address data scarcity in ADD, various augmentation strategies have been explored across text, speech, and multimodal inputs. This study systematically evaluates their effectiveness across multiple datasets, reporting pre- and post-augmentation performance and relative improvements. A summary of methods, datasets, and outcomes is provided in Table IV.

Traditional text-based techniques such as Easy Data Augmentation, Back Translation, and Conditional Data Generation show modest yet consistent gains (1.35%–4.10%), particularly on the Dreaddit dataset. Recent approaches using large language models (LLMs) for synthetic text generation [79] show promise, though their generalizability remains under investigation. In the speech domain, Sub-Dialogue Shuffling [86] yielded a 48.33% improvement on DAIC-WOZ, while Fraug [83] achieved gains

⁵ <https://pan.baidu.com/s/1hXPcjPcqV8ltEi255AjLdQ>

TABLE V
OVERVIEW OF TEXT EMBEDDING TYPES IN ADD SYSTEMS

Embedding Methods	Models
Pre-defined Embeddings	Word2Vec [100], [101], [102], [103], [104], [105] LIWC [103], [106], [107], [108]
DNN-based Embeddings	RNN [109] Transformer [85], [110] BERT [67], [111], [112] BERT, RoBERTa [113] BERT, T5 [114] RoBERTa [115], [116], [86]
Large Language Embeddings	GPT [117], [118], [111], [119], [120] LLaMA [121], [87] Alpala [122]

of 12.65% on CONVERGE and 5.97% on DAIC-WOZ. Topic-based Augmentation [85], a multimodal strategy, improved F1-score from 0.69 to 0.78 by jointly leveraging text and speech.

Speech-based augmentations offer greater performance boosts but exhibit higher variability due to audio complexity. Text-based methods yield smaller, more stable improvements, often influenced by dataset-specific factors. Multimodal techniques demonstrate the most consistent and substantial benefits, underscoring the value of integrating diverse data sources in ADD tasks.

C. Feature Engineering

This section provides a summary of the various feature engineering techniques employed for text and audio data in ADD systems.

1) *Textual Embedding Techniques in ADD Systems*: As shown in Table V, in the early years, pre-defined text embeddings extracted from Word2Vec [92] and LIWC features were commonly used in ADD. Recently, with the advancement of Transformer-based models like BERT [93] and RoBERTa [94] leverage bidirectional encoding and optimized training techniques, producing context-sensitive embeddings that capture sentence-level semantics. These improvements enhance robustness and applicability in NLP tasks. On the other hand, large language models (LLMs) also utilize transformer architectures but differ in their unidirectional encoding approach during pre-training, where each token is conditioned only on its preceding tokens. This structure enables embeddings to capture complex patterns and relationships within the text. Moreover, these embeddings can be fine-tuned on specific downstream tasks using specialized datasets, making them highly adaptable to various applications.

2) *Audio Feature Extraction Methods in ADD Systems*: Table VI summarizes the types of audio features used in ADD, including pre-defined features, high-dimensional embeddings, and hybrid approaches.

Pre-defined features are derived from Acoustic Low-Level Descriptors (LLDs), typically categorized into spectral (e.g., Harmonic Model, Phase Distortion Mean), cepstral (e.g., MFCCs), and prosodic features (e.g., pitch, loudness, rhythm) [80], [95], [96]. Statistical descriptors such as mean, variance, and standard deviation, along with features like Zero

TABLE VI
OVERVIEW OF AUDIO FEATURE EXTRACTION METHODS IN ADD SYSTEMS

Audio Feature Types	Subcategories	Features
Pre-defined features	LLDs	[39], [81], [123], [124], [125], [126], [127], [128], [129]
	Cepstral LLDs	Zero-Crossing Rate [124]
	Prosodic LLDs	Speech Landmarks [87], [101]
	Spectral LLDs	Statistics of LLDs [130], [131]
High-dimensional Features	Others	CNN [82], [132], [133], [96], [116], [110], [85], [134], [135], [136]
	Statistics of LLDs	LSTM [137]
	DNNs	NetVLAD+GRU [75] CNN+LSTM [138]
	SSL Models	Transformers+CNNs [124] Wav2Vec2 [139], [112] Hubert, WavLM, Wav2Vec2 [86] DEPA [140]
Combine Pre-defined and High-dimensional features		MFCC+DNN [80]

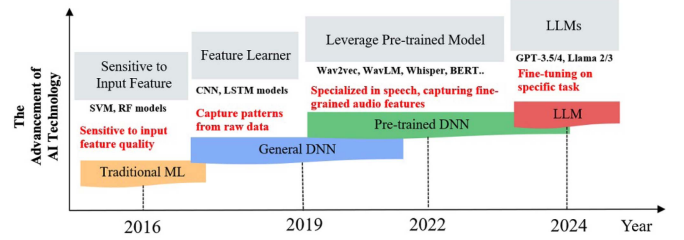


Fig. 6. Evolution of AI techniques in automated depression detection.

Crossing Rate and speech landmarks, are also commonly extracted. These features are often influenced by preprocessing steps such as noise reduction and segmentation, which affect both feature quality and system performance.

In contrast, high-dimensional embeddings generated by deep neural networks capture more abstract temporal and spectral information from spectrogram inputs. Recent advances in self-supervised learning (SSL) have led to the adoption of models like wav2vec 2.0 [97], HuBERT [98], and WavLM [99], which learn robust audio representations from unlabeled data. Their hierarchical architectures enable multi-layer feature extraction, improving accuracy and flexibility in ADD applications.

D. Automated Depression Detection Systems

This section has reviewed the evolution of core architectures in ADD systems across text, audio, and bi-modalities. The following Fig. 6 also proves this trend of methodological innovation in this field.

1) *Text-Based ADD Systems*: Fig. VII summarizes the evolution of text-based ADD systems. Early approaches primarily employed traditional machine learning classifiers such as Support Vector Machines, Random Forests, and Decision Trees [58], [102], [108], [142], [143], [148], [149], but these models struggled to capture complex semantic patterns in language. With the advent of deep learning, researchers began using text embeddings with architectures like Convolutional Neural Networks (CNNs) [84], [103], [106], [145], Long Short-Term Memory (LSTM) networks [150], [151], Graph Attention Network (GAT) [79] and Graph Convolutional Networks (GCN) [152], resulting in improved performance.

TABLE VII
COMPARISON OF TEXT-BASED ADD METHODS

Datasets	Models	Feature Extractor	Classifier	Results
Self-collected	Song et al., 2018 [109]	GloVe	CNN	F1: 0.65
eRISK 2017	Trotzek et al., 2018 [103]	Linguistic metadata	LR	F1: 0.73
Reddit-based	Pirina et al., 2018 [141]	CWINCA	SVM	F1: 0.67
Self-collected	Jain et al., 2019 [142]	NLTK	LR	F1: 0.86
Twitter-based	Zhang et al., 2021	GloVe	XLNet	F1: 0.78
eRISK	Uban et al., 2021 [107]	LIWC	LSTM	F1: 0.45
eRISK T1	MentalRoBERTa [113]	Raw text	MentalRoBERTa	F1: 0.93
Twitter	Chiong et al., 2021 [143]	BOW	LR	F1: 0.90
Dreaddit	Ansari et al., 2021 [76]	R2V	LR	F1: 0.76
Self-collected	Uddin et al., 2022 [144]	LSTM	LSTM	F1: 0.97
RSDD, eRISK, TRT	Nguyen et al., 2022 [145]	PHQ-9 Symptom-Grounded BERT	CNN	F1: 0.85
DAIC-WOZ	Vlad-FRU [75]	Mel spectrogram	BiLSTM	F1: 0.65
DEPTWEET	DistilBERT [59]	BERT	BERT	AUC: 0.80
E-DAIC	DepRoBERTa [115]	Raw text	GPT-3.5 Turbo + DepRoBERTa	F1: 0.82
Dreaddit	Yang et al., 2023 [117]	Raw text	GPT	F1: 0.754
Dreaddit	MentalLLaMA [121]	Raw text	Instruction Finetuned-LLaMA	F1: 0.857
Depression Reddit	Lamichhane et al., 2023 [118]	Raw text	GPT 3.5	F1: 0.86
EATD	Ghosal et al., 2023 [102]	FastText, TF-IDF	XGBoost	F1: 0.71
DSWD	Wang et al., 2023 [77]	Raw text	Medical Knowledge-Augmented LLM	Safety Score: 9.58, Usability Score: 7.74
KID dataset	Ohse et al., 2024 [119]	Raw text	GPT 3.5	F1: 0.82
Dreddit	Mental-Alpaca [122]	Raw text	Instruction Finetuned Alpaca	F1: 0.816
SWDD	Lan et al., 2024 [104]	Raw text	GPT 3.5 Turbo based	F1: 0.76
DAIC-WOZ	Danner et al., [111]	Raw text	Fine-Tuned GPT-3.5 Turbo	F1: 0.78
DAIC-WOZ	Yadav et al., 2023 [146]	RoBERTa-BiLSTM	BiGRU	F1: 0.92
DAIC-WOZ	MDSD-FGPL [114]	T5, BERT	MDSD-FGPL	F1: 0.874
DAIC-WOZ	Hadzic et al., 2024 [120]	Raw text	Instruction Finetuned GPT-4	F1: 0.87
DAIC-WOZ	Burdisso et al., 2024 [147]	GCN	GCN	F1: 0.90

To extract depression-relevant content, two main strategies emerged: topic modeling and reinforcement learning. Topic-based methods [85], [100], [116], [153], [154] often relied on DAIC-WOZ-specific prompts, limiting generalizability. Reinforcement learning, in contrast, enabled data-driven identification of indicative utterances [155].

Recent advancements in pre-trained Transformer models such as BERT and RoBERTa have significantly enhanced classification outcomes. For instance, MentalRoBERTa [113] achieved an F1 score of 0.933 on eRisk, while other variants like DistilBERT [67], Alpaca [122], and DepRoBERTa [115] have shown competitive results across datasets. Notably, Zhang et al. [114] proposed a multi-level classification model combining MDSD-T5 and MDSD-BERT, achieving an F1 of 0.88 on DAIC-WOZ, though the approach faces limitations due to small dataset size and manual prompt design.

More recently, Large Language Models (LLMs) such as LLaMA [156], GPT [157], and Gemini [158] have been fine-tuned using depression-related datasets via techniques like LoRA [159]. Studies report strong performance, with Yang et al. [121] and Ohse et al. [119] fine-tuning LLaMA and GPT-3.5, respectively. Hadzic et al. [120] showed GPT-4 outperforms earlier models without fine-tuning, while Zhang et al. [87] fine-tuned LLaMA2 on DAIC-WOZ with promising results. However, interpretability remains a key challenge for clinical deployment.

While text-based ADD systems have advanced significantly through deep learning and large language models, recent findings caution against over-reliance on contextual signals that may confound diagnostic validity. For example, Li et al., [160] demonstrate that models trained on full clinical transcripts—including interviewer prompts—often achieve higher performance not due to improved inference, but due to shortcut

learning. Interviewer questions can contain implicit cues about the participant’s mental state, allowing models to rely on the phrasing of prompts rather than genuine linguistic markers of depression. Huang et al. [161] further emphasize that failing to distinguish between local (question-level) and global (session-wide) context reduces interpretability and consistency, as the interview structure itself becomes a confounding factor.

Building on this, Guo et al. [116] proposed a topic-aware architecture that explicitly models interviewer–participant dynamics. While this approach improves performance, it also risks amplifying reliance on interviewer phrasing rather than participant-specific signals. These concerns extend beyond clinical interviews. In social media-based ADD, contextual signals such as prior posts or interaction patterns are commonly used to boost performance. Lam et al. [85] show that topic modeling and context-aware augmentation can enhance accuracy, but may also reflect structural artifacts unrelated to mental state. Similarly, Qin et al. [162] found that the performance of large language models varied depending on which parts of interview-style conversations were included, reinforcing that the inclusion or exclusion of context—whether prompts or prior exchanges—can substantially influence model predictions.

These findings indicate the importance of critically evaluating the source and structure of contextual information in ADD tasks. While contextual cues can enhance model accuracy, they also risk introducing bias or masking genuine depressive signals, particularly when models learn to exploit artifacts embedded in interviewer behavior or platform-specific structures.

2) Audio-Based ADD Systems: Fig. VIII provides a quantitative comparison of audio-based ADD systems across datasets, features, classifiers, and performance metrics. Below, we discuss representative models in this domain.

TABLE VIII
COMPARISON OF AUDIO-BASED ADD METHODS

Datasets	Methods	Feature Extractor	Classifier	Results
DAIC-WOZ	DepAudioNet [138]	MFCC	CNN + LSTM	F1: 0.61
DAIC-WOZ	Salekin et al., 2018 [130]	NN2Vec	BLSTM-MIL	F1: 0.8544
DAIC-WOZ	Chlasta et al., 2019 [82]	Raw speech	CNN	F1: 0.57
DAIC-WOZ	Dubagunta et al., 2019 [136]	Raw speech	CNN	F1: 0.67
DAIC-WOZ	Huang et al., 2019 [101]	Acoustic Words and Landmark Words	SVM	F1: 0.667
DAIC-WOZ	Zhao et al., 2020 [137]		BiLSTM	RMSE: 5.66
DAIC-WOZ	Vázquez-Romero et al, 2020 [133]	Mel-spectra	CNN	F1: 0.61
DAIC-WOZ	Muzammel et al., 2020 [96]	LLDs	CNN-based	Acc: 0.86
DAIC-WOZ	Saidi et al, 2020 [132]	Raw speech	CNN + SVM	F1: 0.69
DAIC-WOZ	Jayawardena et al., 2020 [148]	MFCC	LR	RMSE: 6.30
DAIC-WOZ	Yalamanchili et al, 2020 [149]	COVAREP	SVM	F1: 0.84
DAIC-WOZ	Othmani et al., 2021 [80]	Raw speech + MFCC	CNN	F1: 0.655
Self-collected	Kwon et al, 2021 [126]	LLDs	MLP-based	Acc: 0.78
DAIC-WOZ	Rejaibi et al, 2022 [81]	MFCC	RNN	F1: 65.5
Self-collected	Lin et al., 2022 [139]	Wav2Vec	MLP	AUC: 0.868
DAIC-WOZ	Kaur et al, 2022 [163]	spectral, temporal and spectro-temporal	LDA	F1: 0.889
DAIC-WOZ	Speechformer [164]		Speechformer	F1: 0.694
DAIC-WOZ	Vlad-GRU [75]	Mel spectrogram	GRU	F1: 0.66
DAIC-WOZ	CNN AE [165]	Raw speech	SVM	F1: 0.70
DAIC-WOZ	IDL [166]	mel-spectrograms	DepAudioNet	F1: 0.5167
Self-collected	Kim et al., 2023 [83]	Mel spectrograms	CNN	Acc: 0.7814
DAIC-WOZ	Yin et al., 2023 [124]	MFCC	Transformer-CNN-CNN	F1: 0.963
Self-collected	ABAFnet [125]	Upper Envelope, Spectrogram, Mel-spectrogram and HSFs	CNN + LSTM+ attention-based	F1: 0.70
DAIC-WOZ	SFTN [167]	Raw Speech	SFTN	F1: 0.76
E-DAIC	SFTN [167]	Raw Speech	SFTN	RMSE: 5.56
DAIC-WOZ	DALF [168]	MFbanks	DALF	F1: 0.784
DAIC-WOZ	Speechformer-CTC [169]	Wav2Vec	Speechformer	F1: 0.8315

Early work by Ma et al. [138] introduced DepAudioNet, combining CNN and LSTM architectures with a random sampling strategy to mitigate class imbalance. Evaluated on the DAIC-WOZ dataset, it achieved an F1 score of 0.61 and established a baseline for later research. Subsequent studies employed CNN [80], [82], [132], [133], [136], [163], LSTM [96], [130], [137] to process the pre-defined acoustic features or extract high-dimensional features from raw speech. Wang et al. [163] proposed an ECAPA-TDNN-based system leveraging MFCCs and attention mechanisms, achieving 90.8% F1 on clinical speech data. Similarly, Muzammel et al. [96] analyzed consonant and vowel spaces, finding that their fusion improved accuracy to 0.86.

With the rise of self-supervised learning (SSL), models like DEPA [140] and CAE-based systems [165] demonstrated effective unsupervised feature extraction for depression detection. DEPA used a convolutional encoder-decoder to summarize long speech sequences into informative embeddings, while Sardari et al. achieved an F1 score of 0.70 using a CAE-SVM pipeline.

Pretrained Transformer-based models such as wav2vec 2.0 [97], HuBERT [98], and WavLM [99] have recently been adopted for audio-based ADD. Lin et al. [139] fine-tuned these models to achieve an AUC of 0.868. Wu et al. [86] conducted block-wise evaluations, finding WavLM to outperform other SSL models, and introduced Sub-dialogue Shuffling (SDS) for data augmentation, leading to a 78% performance boost. Dumpla et al. [170] further improved results by averaging and concatenating SSL embeddings from multiple models and using a CNN for classification, demonstrating the superiority of SSL-based features over traditional ones such as COVAREP [171].

3) Bimodal ADD Systems: Fig. IX summarizes bimodal ADD systems from 2018 to 2024, which leverage the complementary nature of audio and text modalities. Most studies follow a standard pipeline: extracting acoustic and linguistic features separately before fusing them—either at the feature (early) or decision (late) level. While early fusion integrates modalities before modeling, late fusion—more prevalent in recent works—combines independently processed features, often enhanced via attention mechanisms [112], [116], [173], [174], [175].

Despite its conceptual simplicity, naive concatenation of multimodal features often leads to suboptimal performance due to modality imbalance and noise redundancy. Consequently, more recent research has shifted toward structured fusion techniques using deep learning. Alhanai et al. [131] and Lin et al. [135] demonstrated that fusing text and audio features via LSTM-based architectures significantly outperforms unimodal baselines. Ye et al. [110] achieved an F1 score of 0.906 using DeepSpectrum and word vectors, demonstrating the benefits of well-aligned fusion strategies. Marriwala et al. [134] further explored hybrid fusion pipelines combining CNNs and BiLSTMs for binary depression classification. Transformer-based models have since elevated the performance ceiling of bimodal ADD. Lam et al. [85] used a pre-trained Transformer for text and a CNN for audio, achieving an F1 of 0.87. Toto et al. [112] combined BERT and pre-trained acoustic models (e.g., VGGish, Wav2vec), showing improved performance with demographic features included—though such personalization raises concerns about model generalizability. Guo et al. [116] and Wu et al. [86] adopted attention-based fusion with RoBERTa and WavLM, reaching F1 scores of 0.65 and 0.83 respectively. These studies

TABLE IX
COMPARISON OF BIMODAL ADD METHODS

Datasets	Methods	Text Feature	Audio Feature	Classifier	Results
DAIC-WOZ	Alhanai et al., 2018 [131]	Doc2Vec	COVAREP	Multimodal LSTM	F1: 0.77
DAIC-WOZ	Trf+CNN-Augm [85]	Raw text	Mel spectrogram	Trf+CNN-Augm	F1: 0.87
DAIC-WOZ	Lin et al., [135]	Sentence Embeddings	Mel spectrograms	CNN-BLSTM	F1: 85
DAIC-WOZ	Ye et al., 2021 [110]	Word2Vec	Mel spectrogram	TCN-Trf	F1: 0.906
DAIC-WOZ	AudiBERT [112]	BERT	Wav2Vec	Trf-LSTM	F1: 0.71
DAIC-WOZ	Guo et al., 2022 [116]	RoBERTa	Raw speech	Topic Attentive Trf-based	F1: 64.7
DAIC-WOZ	Park et al., 2022 [172]	BERT	Mel spectrogram	BERT-CNN + CNN-BiLSTM	Acc: 0.90
DAIC-WOZ	Vlad-GRU [75]	ELMo	Mel spectrogram	LSTM-NetVLAD-GRU	F1: 0.71
DAIC-WOZ	Marriwala et al., 2023 [134]	Word2Vec	Mel spectrogram	CNN-LSTM	F1: 0.78
DAIC-WOZ	Wu et al., 2023 [86]	RoBERTa	WavLM	Transformer	F1: 0.83
DAIC-WOZ	Zhang et al., 2024 [87]	Raw text	Speech landmark	Instruction finetuning on Llama2	F1: 0.833

underline the growing importance of leveraging pre-trained SSL models for both modalities and applying attention to better align features temporally and semantically.

In sum, while bimodal ADD systems have shown substantial promise, key limitations remain: reliance on handcrafted fusion strategies, limited generalizability across datasets, and insufficient interpretability. Future research should prioritize unified architectures capable of cross-modal alignment, domain adaptation, and explainability, especially for clinical deployment.

V. LIMITATIONS AND FUTURE DIRECTION

This review systematically examined the current state of ADD systems, highlighting advancements and challenges in utilizing text and audio data for mental health diagnosis. While machine learning and deep learning techniques have shown promising results, several critical areas require further exploration to ensure these technologies can be effectively integrated into clinical practice.

A. Clinical Applicability and Integration With Health Information Systems

Despite advances in ADD, integrating these systems into clinical workflows remains a significant challenge. A primary barrier is interoperability with existing health infrastructure, particularly Electronic Health Records (EHRs) and Clinical Decision Support Systems (CDSS), which often require substantial technical resources and institutional alignment [176]. Moreover, the lack of standardized APIs across EHR platforms demands custom integration solutions that also comply with privacy regulations such as HIPAA [11].

A persistent gap exists between research prototypes and real-world applications. Many ADD models are developed under controlled conditions using limited datasets, which undermines generalizability and performance across diverse populations [177]. Additionally, inconsistencies in depression rating scales used for model training hinder clinical reliability and acceptance [178], [179], [180].

Interpretability remains a further barrier. While deep learning models have improved in this regard, opacity still limits clinical trust. Techniques such as SHAP and LIME have shown promise in making ADD predictions more transparent to clinicians [181], [182], [183]. Transformer-based models like BERT and GPT,

when paired with interpretability tools, offer potential for ethical and trustworthy integration into clinical decision-making processes [111].

B. Data Quality and Cross-Cultural Generalizability

One of the significant challenges in ADD is the variability in data quality, especially with text data sourced from social media or self-reported platforms. These datasets often suffer from noise, biases, and inconsistencies, which can affect model performance.

More importantly, the lack of cross-cultural generalizability limits the broader applicability of ADD systems. Prior studies have shown that emotional expressions—both in semantic content and prosody—differ significantly across languages and cultures [184], [185], [186].

Recent advances in multilingual speech models, such as Wav2Vec 2.0 XLSR [97] and Whisper [187], have demonstrated that pretraining on diverse multilingual datasets can produce language-agnostic acoustic representations. These models have demonstrated promising cross-lingual transferability, especially when lightly fine-tuned on target-language data. However, in clinical applications like ADD, consistent performance still depends on language- and culture-specific validation, given the subtle and diverse ways depression is expressed across populations. Building robust ADD systems may require combining multilingual pretraining with targeted fine-tuning and validation within each cultural context.

C. Longitudinal Data for Better Diagnostic Precision

Most current ADD systems rely on data collected at a single point in time, which provides only a snapshot of a patient's mental state. However, depression is a chronic and fluctuating condition, and its diagnosis and treatment require continuous monitoring [188]. The incorporation of longitudinal data would allow ADD systems to track changes in a patient's mental health over time, providing a more accurate and dynamic assessment. Harrigan et al. provided several recommendations for improving mental health datasets that are curated using self-disclosed diagnoses [[189]]. They suggested that diagnosis dates and psychiatric comorbidities should be annotated to enhance the accuracy of the data. Additionally, they emphasized the importance of sampling control groups using propensity score matching

to ensure comparability. Lastly, they advised identifying and removing spurious correlations that may arise due to selection bias in the dataset [189]. This approach could be particularly useful in remote health monitoring systems and telemedicine, where real-time feedback and adjustments to treatment plans are essential. Future studies should focus on developing methods for collecting and analyzing longitudinal data to improve diagnostic precision and personalized treatment recommendations.

D. Data Augmentation and Enhancing Model Robustness

Data scarcity, especially in multimodal ADD systems, remains a significant limitation. Advanced data augmentation techniques, such as conditional BERT contextual augmentation [190], voice conversion augmentation [191], [192], [193] and back-translation [76] for textual data, have the potential to address this issue by increasing dataset diversity. Additionally, augmentation methods like voice conversion or sub-dialogue shuffling could improve the robustness of audio-based models. These techniques not only enhance the generalization capabilities of ADD systems but also help mitigate the challenges posed by imbalanced datasets. Implementing these strategies in future studies is essential for improving model performance in diverse and real-world scenarios.

E. Ethical and Privacy Considerations

The use of ADD systems in healthcare raises serious ethical and privacy issues, particularly around handling sensitive patient data [40]. Ensuring informed consent and complying with privacy regulations are essential to maintain trust and avoid misuse. Oladosu et al. [194] advocate for privacy-preserving technologies like differential privacy and federated learning, which reduce reliance on centralized data storage and lower the risk of data breaches. Chancellor et al. [195] emphasize that consent should also account for community-level impacts, especially when using data from social media.

Algorithmic bias presents another challenge, particularly when models are trained on imbalanced datasets. To address this, Zuo and Mak [196] developed the Speaker-Invariant Depression Detector (SIDD), which uses latent-space projections and autoencoders to reduce identity-related voice biases. Cheong et al. [197] propose causal multimodal fusion and referee-guided tuning to improve both group and individual fairness. Bailey and Plumbley [198] show that using raw audio features—rather than Mel-spectrograms—helps mitigate gender imbalance, especially when combined with data redistribution techniques. Additionally, interpretability tools like SHAP can help identify and mitigate biased training inputs [199].

For ADD systems to be ethically viable in clinical contexts, future research must prioritize fairness, transparency, and privacy-preserving machine learning frameworks [200].

VI. CONCLUSION

This review examined the current state of Automated Depression Detection (ADD) systems using text and audio data,

highlighting their potential to improve mental health diagnostics and ease the burden on clinicians. Despite progress, challenges remain, including data quality, cultural bias, limited interpretability, and the lack of longitudinal data.

A practical starting point for new research or system development in this field is the selection of a suitable dataset, such as DAIC-WOZ and MODMA. To establish a strong foundation, researchers should begin with existing state-of-the-art open-source systems to create a baseline. Initial implementation can focus on unimodal modeling (e.g., text or audio only), gradually expanding to bimodal or fully multimodal architectures as the system matures.

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